Daniel Park

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Education

Western University — Sept. 2021 – Apr. 2026
Specialization in ***Computer Science***

- Member of **Western AI**, **Western CS Undergraduate Society** and Events Assistant at Western Sneaker Club.

Relevant Courses: Machine Learning, Databases, Statistical Programming, Unstructured Data, Operating Systems, Data Structures and Algorithms, Computer Networks.

Professional Experiences

Data & Machine Learning Intern — Sept. 2025 – Apr. 2026
HEAL

- Architected an automated, cloud-native **ETL pipeline** on **AWS** utilizing **Apache Airflow** to orchestrate data ingestion from the **Google Places API**, detecting spatial anomalies and ensuring cross-system integrity for downstream **machine learning (ML)** applications.
- Developed robust **data transformation** micro-workflows in **Python (pandas, NumPy, regex)** to sanitize and reconcile unstructured text data, significantly reducing anomalies and decoupling data preparation from analytics flows.
- Designed a scalable, version-controlled **data model** and storage architecture on **AWS** with rigorous schema validation, establishing centralized governance protocols that guaranteed high availability and long-term pipeline reliability.
- Deployed predictive **machine learning models** utilizing **Scikit-Learn** and **XGBoost** (clustering, time-series forecasting) to analyze retail density trends, translating complex data into actionable visualizations to drive **strategic, data-driven decisions**.
- Implemented **MLOps** best practices for model validation and performance monitoring, collaborating with **cross-functional stakeholders** to translate complex analytical outputs into clear technical specifications and business requirements.

Data Science Intern — Sept. 2024 – Aug. 2025
interRAI

- Managed the end-to-end **ETL** and data preparation lifecycle for high-volume clinical datasets using **PySpark**, optimizing distributed validation processes to ensure high-fidelity inputs for advanced health research.
- **Modernized** legacy data processing flows into automated, distributed workloads utilizing **Spark SQL**, reducing manual processing overhead and accelerating execution runtimes by approximately **30%** for large-scale analysis.
- Spearheaded a comprehensive technical training initiative and documented core **Spark** architectures, accelerating team onboarding and enabling unified **cross-functional collaboration** across multidisciplinary research units.

Front End Developer Intern — May 2024 – Aug. 2024
BERL

- Architected highly responsive front-end applications by refactoring legacy **React** layouts, utilizing **CSS Flexbox**/**Grid**, and extending **Cascade CMS** templates, significantly reducing system latency and streamlining digital content delivery.
- Transformed monolithic administrative interfaces into streamlined operational workflows while enforcing strict accessibility and code quality standards through rigorous peer code reviews, ensuring **continuous improvement**, high-performance UI deployments, and system stability.

Software Developer Intern — Sept 2023 – Apr. 2024
Cultureplex

- Engineered custom **Python (pandas, NumPy)** and **SQL**-based analytical frameworks to process and aggregate complex datasets, accelerating large-scale data extraction for digital humanities research initiatives.
- Optimized **PostgreSQL** database architecture through advanced **data modeling**, schema normalization, and strategic query tuning, mitigating performance bottlenecks and significantly reducing data retrieval latency.
- Introduced a novel **ETL pipeline** leveraging **Python** and **LLMs (Generative AI)** to systematically **transform** unstructured text into structured formats, enabling seamless tool adoption by **cross-functional** research teams.
- Generated high-impact visualizations (**Matplotlib**/**Seaborn**), translating complex aggregated data into unified **Excel**/**CSV** structures to support stakeholders in accelerating academic publications.

Projects | View All

Canadian Ski Resorts Powder Predictor | *Airflow, PySpark, AWS (S3/RDS), XGBoost, Flask, React*

- Architected an end-to-end **MLOps pipeline** using **Apache Airflow** to automate daily ingestion of 7-day weather forecasts for 18 Canadian ski resorts via **Open-Meteo REST APIs**.
- Built a cost-effective **data lake on AWS S3** for raw storage and implemented a **PySpark ETL** job to compute complex meteorological features, including 3-day rolling snowfall and freeze-thaw cycles.
- Trained an **XGBoost Regressor** on 62,000+ historical records to predict a "Powder Score" (1-10), achieving an MAE of 0.01 and deploying model inference to an **AWS RDS PostgreSQL** warehouse.
- Developed a **Flask REST API** to serve insights to a dark-mode **React dashboard**, enabling users to filter and identify optimal skiing conditions.

Technical Skills & Certifications

Data Science & ML: Python, R, PyTorch, TensorFlow, Scikit-Learn, XGBoost, LLMs / Generative AI, Pandas, NumPy, Polars, SciPy, PySpark, Matplotlib, Seaborn, SHAP, UMAP, Yellowbrick, Jupyter
Languages & Web: SQL, Java, C, C++, C#, Bash, PHP, HTML/CSS, XML, Django, .NET, React
Cloud, Data Engineering & DevOps: ETL / ELT, Data Pipelines, Data Modeling, Airflow, AWS, Azure, GCP, Docker, Git, Maven, Elasticsearch, Lucene, MapReduce, Regex, Excel, CSV, Claude Code
Databases: MySQL, DynamoDB, Snowflake, Hadoop, Firebase, MongoDB, SQLite, PostgreSQL, MSSQL

- **AWS** Machine Learning Engineer Certificate - In Progress
- **Meta** Backend Developer Specialization Certificate